CHEST DISEASES HOSPITAL | LUNG TRANSPLANT UNIT

Post-operative Lung Transplant Orders

Cardiothoracic ICU — Immediate Post-operative Order Set

Ref: CDH-LTU-PPO-03
Version 1.0 | April 2026

Patient Name:	MRN / Civil ID:	Date:	Time:
Weight (kg):	**Allergies:**	**Procedure Type:**	**Surgeon:**

Procedure	☐ ☐ Right Single Lung Transplant ☐ ☐ Left Single Lung Transplant ☐ ☐ Bilateral Sequential Lung Transplant
ECMO/CPB Used	☐ ☐ No bypass used ☐ ☐ VA-ECMO (intraoperative; decannulated) ☐ ☐ VA-ECMO continuing to ICU ☐ ☐ CPB used

1. ADMISSION & ISOLATION

☐	**Admit to: Cardiothoracic ICU — STRICT ISOLATION room. Single-room isolation mandatory.**
☐	Notify: Transplant Coordinator, Transplant Physician on call, Respiratory Therapy, Physiotherapy.
☐	Diagnosis: Lung Transplant — Post-operative Day 0. Refer to CDH-LTU-PERIOP-02 for full pathway.

2. HAEMODYNAMICS & FLUID MANAGEMENT (KEEP DRY)

CRITICAL Target negative fluid balance. CVP <8 mmHg. Prefer vasopressors over fluid boluses for MAP support. Avoid fluid overload — directly worsens Primary Graft Dysfunction.

☐	**Vital Signs & Monitoring**	☐	Continuous HR, Art-line BP, CVP, SpO_2. Q1h neuro checks. Temperature q4h.

Haemodynamic Target	Goal	Intervention
Mean Arterial Pressure	65–80 mmHg	☐ Norepinephrine infusion (first-line). Start at 0.05 µg/kg/min; titrate. ☐ Epinephrine if refractory RV failure (per ICU physician).
Central Venous Pressure	<8 mmHg	☐ D5W at 30 mL/hr to KVO (maintenance). ☐ Furosemide (Lasix) _____ mg IV q _____ hrs.

3. RESPIRATORY & VENTILATOR MANAGEMENT

Parameter	Mode	TV (mL/kg IBW)	PEEP (cmH_2O)	FiO_2
Initial Settings		6 mL/kg	5–8	Lowest to maintain SpO_2 ≥94%

☐	☐ ABG q4h AND 30 minutes after any ventilator change.
☐	☐ Aggressive pulmonary toilet: in-line suctioning PRN. Chest physiotherapy q4h while intubated.
☐	☐ Extubation: wean per ICU protocol when awake, haemodynamically stable, and ABG acceptable. Target within 12–24 hours.
☐	☐ Post-extubation: incentive spirometry Q1H while awake.

☐	☐ Bronchoscopy: routine bedside bronchoscopy Day 1 post-operatively (inspect anastomoses, clear secretions). Document findings.

4. IMMUNOSUPPRESSION — CRITICAL TIMING

CRITICAL Immunosuppression timing is CRITICAL. Administer all agents at specified times. Document batch, dose, route, and exact time for every immunosuppressant administered.

Agent	Order
Induction Agent	☐ ☐ Basiliximab 20 mg IV on POD 0 AND POD 4 (IL-2 receptor antagonist) ☐ ☐ Alemtuzumab 30 mg IV on POD 0 (pre-medicate: antihistamine + hydrocortisone) — as directed by transplant physician
Corticosteroids	☐ Methylprednisolone _____ mg IV q8h (intraoperative first dose given at reperfusion — see Anaesthesia record). Taper per transplant physician protocol.
Calcineurin Inhibitor	☐ Tacrolimus _____ mg PO/NG/SL BID. ☐ Cyclosporin _____ mg PO/NG BID (if tacrolimus intolerant). **Check 12-hour trough level DAILY at 06:00. Target trough: _____ ng/mL (Month 1–3: 10–15 ng/mL; Month 4–12: 8–12 ng/mL).**
Antimetabolite	☐ Mycophenolate Mofetil (MMF / CellCept) 1000 mg IV/PO BID.

5. INFECTION PROPHYLAXIS

PROTOCOL Targeted to donor/recipient cultures. Transplant Infectious Disease consultation mandatory within 24 hours of admission.

Category	Agent	Dose / Duration / Notes
Bacterial	☐ Piperacillin/Tazobactam ☐ Meropenem ☐ Vancomycin ☐ Other: _____	Broad-spectrum; tailored to donor/recipient airway cultures within 48–72 hours. _____ mg IV q _____ hrs.
Fungal	☐ Voriconazole 200 mg IV BID ☐ Inhaled Amphotericin B ☐ Other: _____	Duration per protocol; adjust per culture results and galactomannan monitoring.
CMV Antiviral	☐ Valganciclovir 900 mg PO daily ☐ IV Ganciclovir 5 mg/kg daily (if NPO/NG)	**CMV serostatus: D _____ / R _____** ☐ D+/R−: 6 months minimum. ☐ D+/R+: 3 months minimum. ☐ D−/R−: Acyclovir only.
PCP	Trimethoprim/Sulfamethoxazole (TMP-SMX) SS 1 tab PO daily	Lifelong prophylaxis. Dose-adjust for renal function.

6. DRAINS, TUBES & GI PROPHYLAXIS

Chest Tubes	☐ ☐ Right pleural ☐ Left pleural ☐ Mediastinal ☐ Maintain on −20 cmH_2O suction initially. ☐ Transition to water seal when drainage serous and <200 mL/24h with no air leak. ☐ NOTIFY surgeon if bleeding >150 mL/hr for 2 consecutive hours.
Urinary Catheter	☐ Foley catheter to gravity. Hourly urine output monitoring. NOTIFY if UOP <0.5 mL/kg/hr.
Nasogastric / Dobhoff	☐ Insert NGT/Dobhoff for feeds. Low continuous suction if not feeding. Do NOT commence enteral feeds until extubated and swallow assessment completed.

GI Prophylaxis	☐ Pantoprazole 40 mg IV daily (proton pump inhibitor — stress ulcer prophylaxis).

7. ANALGESIA

☐	☐ Thoracic epidural (preferred): manage per Anaesthesia Pain Service. Document level and current infusion.	☐	☐ OR Bilateral ESPB: if anticoagulated; manage per Anaesthesia Pain Service.
☐	☐ Paracetamol 1 g IV q6h SCHEDULED.	☐	☐ Fentanyl PCA (if no epidural/ESPB).
☐	☐ NO NSAIDs: Nephrotoxic in combination with CNIs (tacrolimus/cyclosporin). ABSOLUTE CONTRAINDICATION.	☐	☐ Gabapentin 100 mg PO BID may be added for neuropathic pain (with physician review).

8. LABORATORY INVESTIGATIONS

☐	Daily Morning Labs (04:00–06:00): CBC with differential, CMP (Na, K, Cr, BUN, LFTs, Albumin), Mg, Phos, PT/INR, PTT.	☐	Tacrolimus trough level drawn exactly at 06:00 (12-hour post-last dose). Target per immunosuppression orders.
☐	Portable CXR: daily every morning AND immediately post-extubation. Radiology to read same day.	☐	12-lead ECG: daily ×3 days post-operatively.
☐	ABG q4h AND 30 minutes after any ventilator change.	☐	CMV PCR at Day 7, then weekly ×4 weeks, then monthly to 12 months (see CDH-LTU-PERIOP-02 Section 3.3).

9. DVT PROPHYLAXIS

☐	☐ Sequential Compression Devices (SCDs): bilaterally at all times when in bed, from admission.
☐	☐ Heparin 5000 units SC q8h: START POD 1 when chest tube drainage <50 mL/hr. HOLD if active bleeding.

10. PHYSIOTHERAPY & EARLY MOBILISATION

Phase	Order
While intubated	☐ Chest percussion and vibration q4h. ☐ Passive limb exercises. ☐ Head-of-bed elevation 30–45°.
POD 1 (post-extubation)	☐ Out of bed to chair if haemodynamically stable (MAP ≥65 mmHg, SpO_2 ≥90% on current O_2). ☐ Incentive spirometry every hour while awake. ☐ Walking with physiotherapist to bedside.
POD 2–3	☐ Hallway ambulation with continuous SpO_2 monitoring. ☐ Progressive distance increase as tolerated. ☐ Stationary cycling if available and tolerated.

11. ADDITIONAL CLINICAL ORDERS

Blood Glucose	☐ Blood glucose q4h initially. Insulin sliding scale per ICU protocol. Target glucose 6–10 mmol/L (steroid-induced hyperglycaemia common post-transplant).
Electrolytes	☐ Replace Mg and Phos to normal range. Tacrolimus causes Mg wasting — repletion is critical. Replace K to maintain K ≥4.0 mmol/L.
Transplant ID Consult	☐ Infectious disease consultation within 24 hours for antimicrobial guidance based on donor/recipient cultures.
Dietitian	☐ Dietitian consult within 24 hours. Immunocompromised diet. Grapefruit/pomelo restriction (CYP3A4 inhibition with tacrolimus). Target 25–30 kcal/kg/day.
Social Work	☐ Social work consult to assess caregiver support and discharge planning from Day 3 onwards.

PRESCRIBER VERIFICATION & SIGNATURE		
Prescribing Physician	**Pharmacy Verification**	**Nursing Check**
Name:	**Pharmacist Name:**	**RN Name:**
Signature:	**Signature:**	**Signature:**
Date & Time:	**Date & Time:**	**Date & Time:**

ORDER VALIDITY These orders are valid for a maximum of 24 hours from the time of writing. Review and rewrite daily or with any significant clinical change. Any modification must be signed, dated, and timed by the prescribing physician.